A FINITE PURELY INSEPARABLE EXTENSION MAKES THE REDUCTION OF A FINITE TYPE SCHEME GEOMETRICALLY REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex, lines 1-77), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference, slogan and history display environments; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the four cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. Pre-open checks run for this card: (1) same-name intersection filter across staged chapters (no hit); (2) cross-lane scan over every card in focused-seeds and the preparation worklist via agent-authoring/stk/card_plan.py –scan (no card claims this label).

Allowed standard prerequisites (editorial).

- A purely inseparable extension does not change the underlying topological space of the reduction. Algebra, Lemma labelled algebra-lemma-p-ring-map (canonical Stacks tag 0BRA); invoked at proof lines 651-653. Supplies that (X_{k'})_{red} -> X is a homeomorphism, so the points over the generic points of the components are again the generic points.
- Local rings at the generic points of a reduced scheme are fields. Algebra, Lemma labelled algebra-lemma-minimal-prime-reduced-ring (canonical Stacks tag 00EU); invoked at proof lines 656-657. Supplies that K_i = O_{X,x_i} is a field, and (with local finiteness of type) that K_i/k is finitely generated.
- Making finitely generated field extensions separable by a finite purely inseparable base change. Algebra, Lemma labelled algebra-lemma-make-separable (canonical Stacks tag 030R); invoked at proof lines 661-664. Supplies the single finite purely inseparable k'/k that makes all (K_i tensor_k k')_{red} separable over k', hence geometrically reduced by algebra-lemma-characterize-separable-field-extensions (030W, line 667).
- Geometric reducedness is detected at the generic points of the irreducible components. Same chapter, Lemma labelled lemma-generic-points-geometrically-reduced (source line 555, Stacks tag 04KS), part (3); invoked at proof lines 668-670. Supplies the final step from the local rings at the generic points to geometric reducedness of the whole scheme.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0BRA: sources/stacks-chow/tags.txt, source lines 14763–14763. Cross-chapter standard prerequisite invoked once at proof line 653 for the homeomorphism (X_{k'})_{red} -> X; the generic-point bookkeeping on top of it is retained in the counted span.

Source reference. Original lemma 00EU: sources/stacks-chow/tags.txt, source lines 548–548. Cross-chapter standard prerequisite invoked once at proof lines 656-657 to see that the local rings at generic points of a reduced scheme are fields.

Source reference. Original lemma 030R: sources/stacks-chow/tags.txt, source lines 3958–3958. Cross-chapter standard prerequisite invoked once at proof lines 661-664 for the single finite purely inseparable extension making all the residue fields separable; the rest of the proof is retained in the counted span.

Source reference. Original lemma 030W: sources/stacks-chow/tags.txt, source lines 3963–3963. Cross-chapter standard prerequisite invoked once at proof line 667 for separable implies geometrically reduced over the base field.

Source reference. Original lemma 04KS: sources/stacks-varieties/pinned-original-source.tex, source lines 555–555. Same-chapter pointer invoked at proof lines 668-670, part (3), as the criterion that upgrades the local checks at generic points to geometric reducedness of the scheme; it is the closing step only.

Lemma 0.1. *Let k be a field. Let $X \to \mathrm{Spec}(k)$ be locally of finite type. Assume X has finitely many irreducible components. Then there exists a finite purely inseparable extension k'/k such that $(X_{k'})_{red}$ is geometrically reduced over k'.*

Proof. To prove this lemma we may replace X by its reduction X_{red}. Hence we may assume that X is reduced and locally of finite type over k. Let $x_1, \ldots, x_n \in X$ be the generic points of the irreducible components of X. Note that for every purely inseparable algebraic extension k'/k the morphism $(X_{k'})_{red} \to X$ is a homeomorphism, see Algebra, Lemma 0BRA. Hence the points $x'_1, \ldots, x'_n$ lying over $x_1, \ldots, x_n$ are the generic points of the irreducible components of $(X_{k'})_{red}$. As X is reduced the local rings $K_i = \mathcal{O}_{X,x_i}$ are fields, see Algebra, Lemma 00EU. As X is locally of finite type over k the field extensions K_i/k are finitely generated field extensions. Finally, the local rings $\mathcal{O}_{(X_{k'})_{red},x'_i}$ are the fields $(K_i \otimes_k k')_{red}$. By Algebra, Lemma 030R we can find a finite purely inseparable extension k'/k such that $(K_i \otimes_k k')_{red}$ are separable field extensions of k'. In particular each $(K_i \otimes_k k')_{red}$ is geometrically reduced over k' by Algebra, Lemma 030W. At this point Lemma 04KS part (3) implies that $(X_{k'})_{red}$ is geometrically reduced. □